

Shoot or Keep the Ball? A Held-Out Causal Audit of Shot Selection

Introduction

Every attack in soccer forces a choice: shoot now, or keep the ball by passing, carrying or dribbling. Comparing these alternatives is hard because players shoot when chances already look good. Recent tracking-data models recommend when to shoot, without identified causal effects (Verronneau et al., MLSA 2026). We ask: when players shot, did it pay off compared with what they would typically have done instead?

Methods

We use StatsBomb open event data with 360 freeze-frames (player positions). Decisions are live-play actions in the attacking third with the ball's 3 m surroundings fully visible. Each shot is compared with keeping the ball the way players in the same situation actually did, through the observed mix of passes, carries and dribbles. We compare only where both choices were realistic (94.9% of eligible shots). The outcome is the first goal within 15 seconds (+1 scored, -1 conceded, 0 otherwise). A doubly robust estimator, cross-fitted by match, adjusts for geometry, pressure, score, time and recent actions. The analysis was fixed on Euro 2020 and 2024 (102 matches), then run once on World Cup 2022 (64 matches).

Results

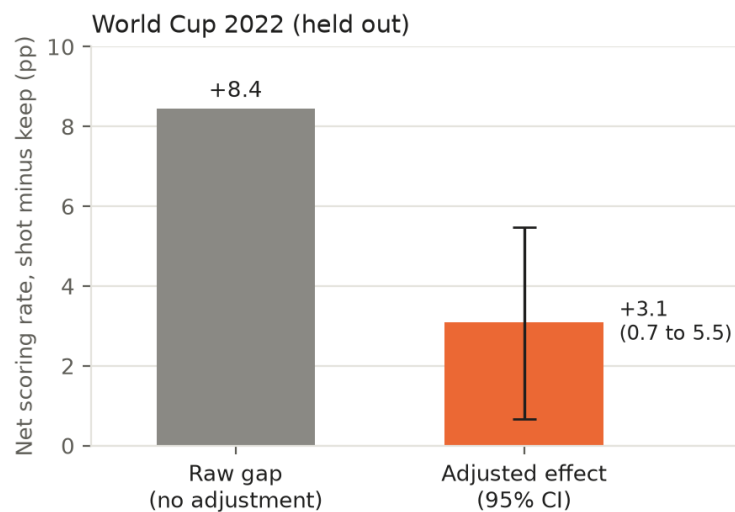
Before weighting, shots and other decisions sit 2.2 standard deviations apart in distance to goal; after, every adjusted covariate differs by less than 0.10. In the World Cup (1,228 shots; 20,532 other decisions), the net scoring rate was 10.7% after shots, against an estimated 7.7% had the same players kept the ball: a gain of 3.1 percentage points (95% CI 0.7 to 5.5, clustered by match), about a third of the raw gap of 8.4 (Figure 1). Point estimates were similar across specifications: outcome models and estimators give 2.8 to 3.4; 5-, 10- and 30-second windows give 3.2 to 3.9; leaving out any one match or team gives 2.3 to 4.1. Randomly reassigned shots show no effect, and in 400 simulations with known effects, intervals contained the truth 93% of the time.

Exploratory, prespecified subgroups place the gain inside 18 m in both tournaments (Figure 2; World Cup +5.9, 2.2 to 9.7), not beyond (-1.3); each narrowly misses one balance threshold, and their difference was untested.

Limits remain. The Euro estimate points the same way but is uncertain (1.8; -0.2 to 3.8). Clustering by opponent widens the interval to include zero (-0.3 to 6.4), and hidden factors shifting shooting odds 1.35-fold would bring the estimated lower sensitivity bound (not the confidence limit) to zero.

Conclusion

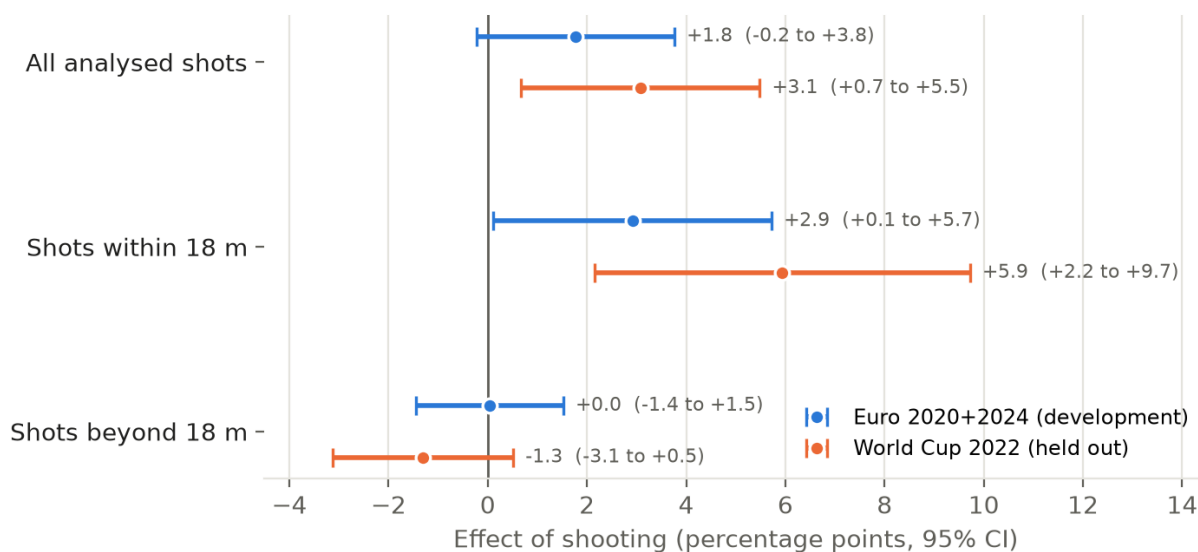
Public event data can answer what expected goals cannot: what shooting added over the realistic alternative. Under stated assumptions, in a tournament held out while the primary analysis was specified, players' shots added about three percentage points of net scoring rate, with stable point estimates across specifications, matches and teams. The result is a rerunnable public audit for testing shot-selection claims, including those of recommendation models.



Adjusted: 10.7% after shots vs 7.7% estimated had the same players kept the ball.

Figure 1: Raw vs adjusted effect, World Cup.

Adjusted effect of shooting vs keeping the ball, overall and by shot distance



Net scoring rate = first goal within 15 s (+1 scored, -1 conceded). Bars: 95% CI, clustered by match. Distance groups are prespecified.

Figure 2: Adjusted effect by distance (exploratory).